Ch 5.1.4-5: More Cross-Validation

Lecture 14 - CMSE 381

Prof. Guanqun Cao

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Oct 1, 2025

Announcements

Last time:

- k-fold CV

This lecture:

- More k-fold CV
- Bias-Variance Tradeoff
- CV for classification

Announcements:

- Exam 1 feedback sent

CMSE381_F2025_Schedule : Schedule

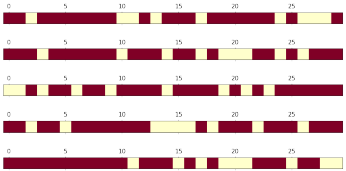
	M	9/22	***Project Day & Review***		
	W	9/24	**Midterm #1**		
12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	

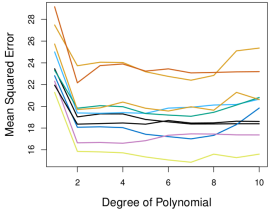
Section 1

k-fold CV

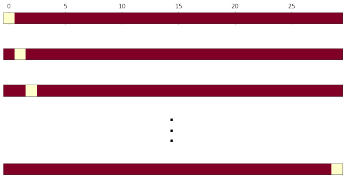
Approximations of Test Error

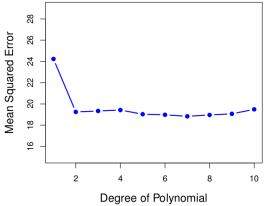
Validation Set



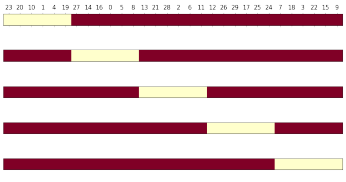


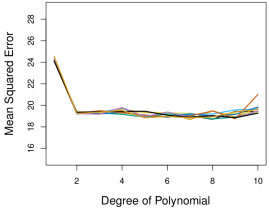
LOOCV





K-fold CV



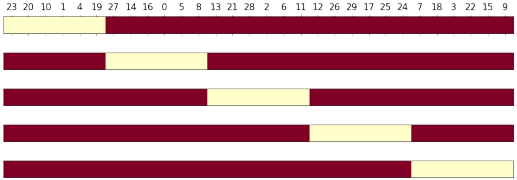


Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

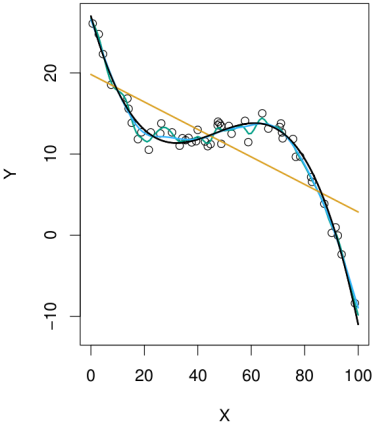
- Rinse and repeat

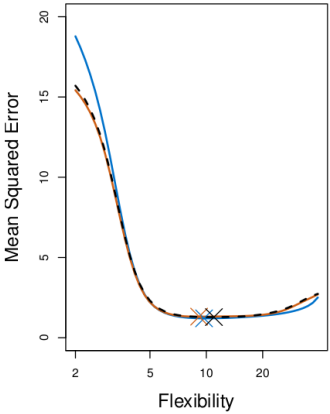


Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

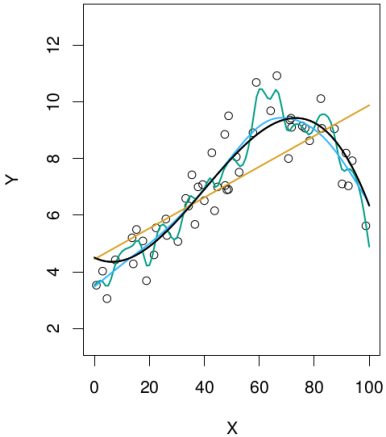
Comparison with simulated data: Ex 3

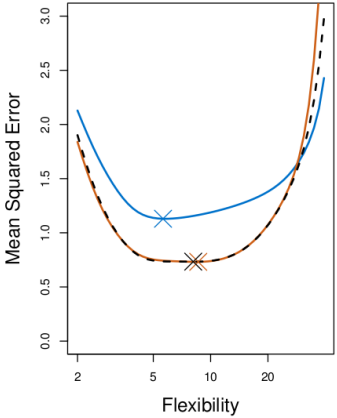




- Blue: true test MSE is shown in blue
- Black dashed: the LOOCV estimate
- Orange: 10-fold CV
- The crosses indicate the minimum of each of the MSE curves.
- Nearly the same

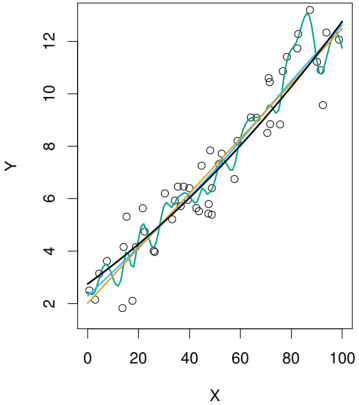
Comparison with simulated data: Ex 1

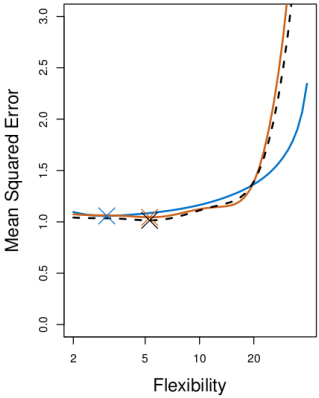




- Blue: true test MSE is shown in blue
- Black dashed: the LOOCV estimate
- Orange: 10-fold CV
- The crosses indicate the minimum of each of the MSE curves.
- **Underestmates the true test MSE**

Comparison with simulated data: Ex 2





Close for lower flexibility, then overestimates at higher ranges

Takeaways from the examples

- In all plots, LOOCV and 10-fold CV are similar
- Sometimes really only care about the minimum point in the test MSE because want to get the right degree of freedom for a model. All examples have approximately same x-coord for that

Bias-Variance Tradeoff: Bias

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

- note we talk about bias in estimation of test error rate
- 1.... Validation set overestimates test error b/c used small subset of data
- 3.... k-fold gives medium level of bias b/c training set has approximately $(k-1)n/k$ observations
- 2....LOOCV gives approximately unbiased estimate since uses almost all data every time
- soooooooooo LOOCV better than k-fold just for bias

Bias-Variance Tradeoff: Variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

- note we talk about variance in estimation of test error rate
- LOOCV higher variance:
  - avg the outputs of n fitted models, but all on almost identical observations
  - high correlation with each other
  - repeated measure problem in science! - don't generalize well across samples
- k-fold
  - k fitte models somewhat less correlated with each other
  - mean of many highly correlated quantities has higher variance than those not corrleated
- Empirically, $k = 5$ or 10 has been shown to be a happy medium

In short: Vadidation vs Test

- all the time, we are pretending the validation set etc is the test set...
- when it is not.
- a true test set is an independent sample
- CV measures will never achieve that

Real-world example: Chekroud et al., Science 383, 164–167 (2024)

RESEARCH ARTICLE

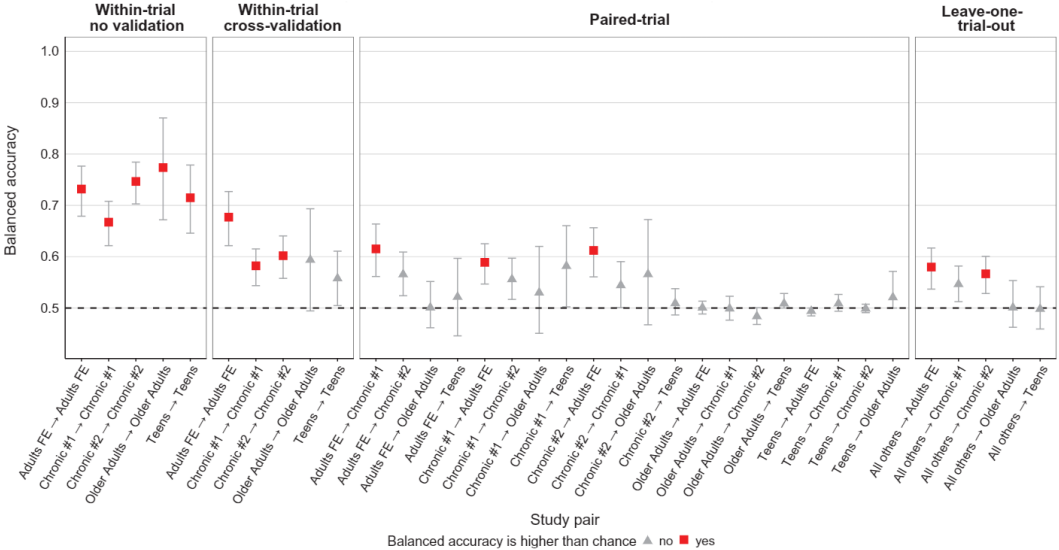
NEUROSCIENCE

Illusory generalizability of clinical prediction models

Adam M. Chekroud[1,2*], Matt Hawrilenko[1], Hieronimus Loho[2], Julia Bondar[1], Ralitza Gueorguieva[3], Alkomiet Hasan[4], Joseph Kambeitz[5], Philip R. Corlett[2], Nikolaos Koutsouleris[6], Harlan M. Krumholz[7], John H. Krystal[2], Martin Paulus[8]

It is widely hoped that statistical models can improve decision-making related to medical treatments. Because of the cost and scarcity of medical outcomes data, this hope is typically based on investigators observing a model's success in one or two datasets or clinical contexts. We scrutinized this optimism by examining how well a machine learning model performed across several independent clinical trials of antipsychotic medication for schizophrenia. Models predicted patient outcomes with high accuracy within the trial in which the model was developed but performed no better than chance when applied out-of-sample. Pooling data across trials to predict outcomes in the trial left out did not improve predictions. These results suggest that models predicting treatment outcomes in schizophrenia are highly context-dependent and may have limited generalizability.

Real-world example: Chekroud et al., Science 383, 164–167 (2024)



Section 2

Using K-Fold CV on Polynomial Linear Regression

Polynomial regression

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \beta_d x_1^d + \varepsilon_i$$

- Can learn with linear models by passing in predictors x_i^ℓ
- Tend to not go higher than degree 3 or 4 because makes overly flexible

Faking linear regression into doing our work for us

- Draw a column of data
- Draw the version with squares and cubes in the columns
- Pass to linear regression and interpret the coefficients as the ones we want.

Coding - Build a plot for train/test scores vs flexibility

Next time

CMSE381_F2025_Schedule : Schedule

	M	9/22	***Project Day & Review***		
	W	9/24	**Midterm #1**		
12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	